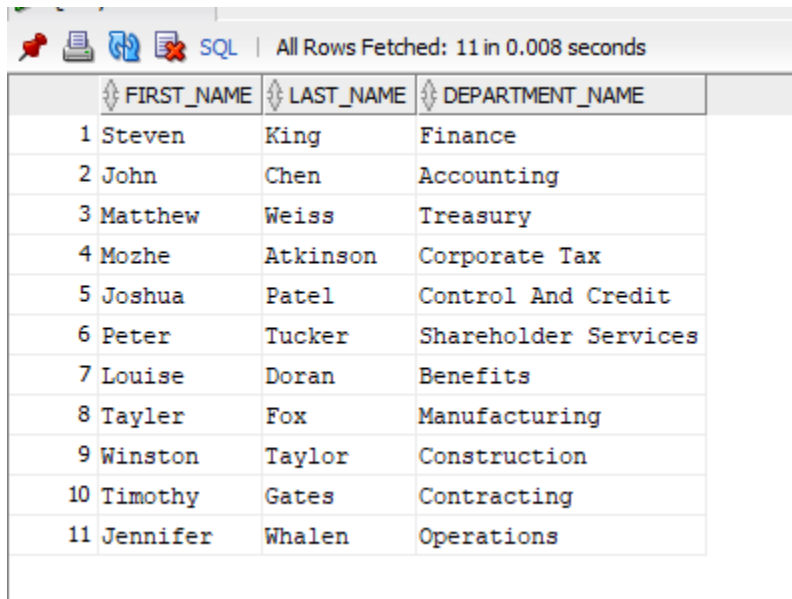


Query Review 1

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
JOIN departments d
ON e.employee_id = d.department_id;
```



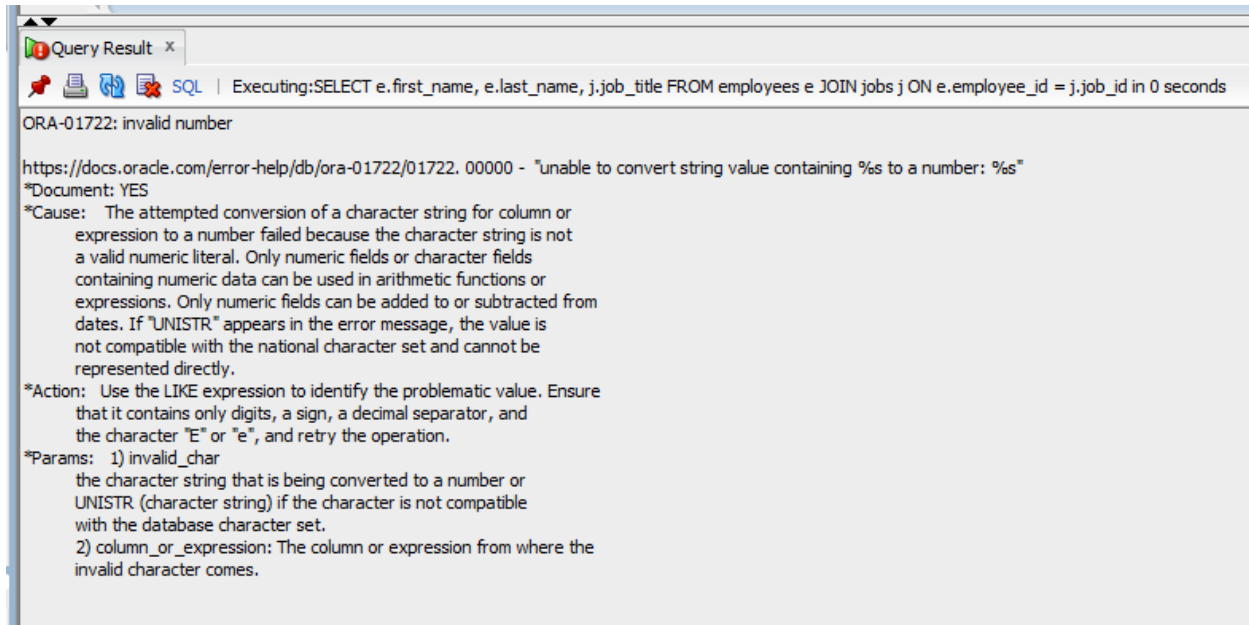
	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Steven	King	Finance
2	John	Chen	Accounting
3	Matthew	Weiss	Treasury
4	Mozhe	Atkinson	Corporate Tax
5	Joshua	Patel	Control And Credit
6	Peter	Tucker	Shareholder Services
7	Louise	Doran	Benefits
8	Taylor	Fox	Manufacturing
9	Winston	Taylor	Construction
10	Timothy	Gates	Contracting
11	Jennifer	Whalen	Operations

The query makes a table with (First_Name) from Employee table, (Last name) from employee table and (department name) from department table and joins them in one table where the employee_id from the Employee table equals the department_id from the department table. It shows output but logically it's wrong; We can not match employee's unique id with departments unique id. To fix the problem:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id;
```

Query Review 2

```
SELECT e.first_name, e.last_name, j.job_title
FROM employees e
JOIN jobs j
ON e.employee_id = j.job_id;
```



in the quesr : ON e.employee_id = j.job_id ,

- **e.employee_id** is a numeric column containing values like 100, 101, 102.
- **j.job_id** is a text column , containing alphanumeric string codes like ' IT_PROG' , 'HR_REP' , or 'AD_PREP' .

How to fix:

```
SELECT e.first_name, e.last_name, j.job_title FROM employees e JOIN jobs j ON e.job_id = j.job_id;
```

Query Result x			
                                			
SQL Fetched 50 rows in 0.007 seconds			
	FIRST_NAME	LAST_NAME	JOB_TITLE
1	William	Gietz	Public Accountant
2	Shelley	Higgins	Accounting Manager
3	Jennifer	Whalen	Administration Assistant
4	Steven	King	President
5	Neena	Kochhar	Administration Vice President
6	Lex	De Haan	Administration Vice President
7	Daniel	Faviet	Accountant
8	Luis	Popp	Accountant
9	John	Chen	Accountant
10	Ismael	Sciarra	Accountant
11	Jose Manuel	Urman	Accountant
12	Nancy	Greenberg	Finance Manager
13	Susan	Mavris	Human Resources Representative
14	Bruce	Ernst	Programmer
15	Alexander	Hunold	Programmer
16	David	Austin	Programmer
17	Diana	Lorentz	Programmer
18	Valli	Pataballa	Programmer
19	Michael	Hartstein	Marketing Manager
20	Pat	Fay	Marketing Representative
21	Hermann	Baer	Public Relations Representative
22	Sigal	Tobias	Purchasing Clerk
23	Shelli	Baida	Purchasing Clerk

Query Review 3

```
SELECT e.first_name, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'Sales';
```



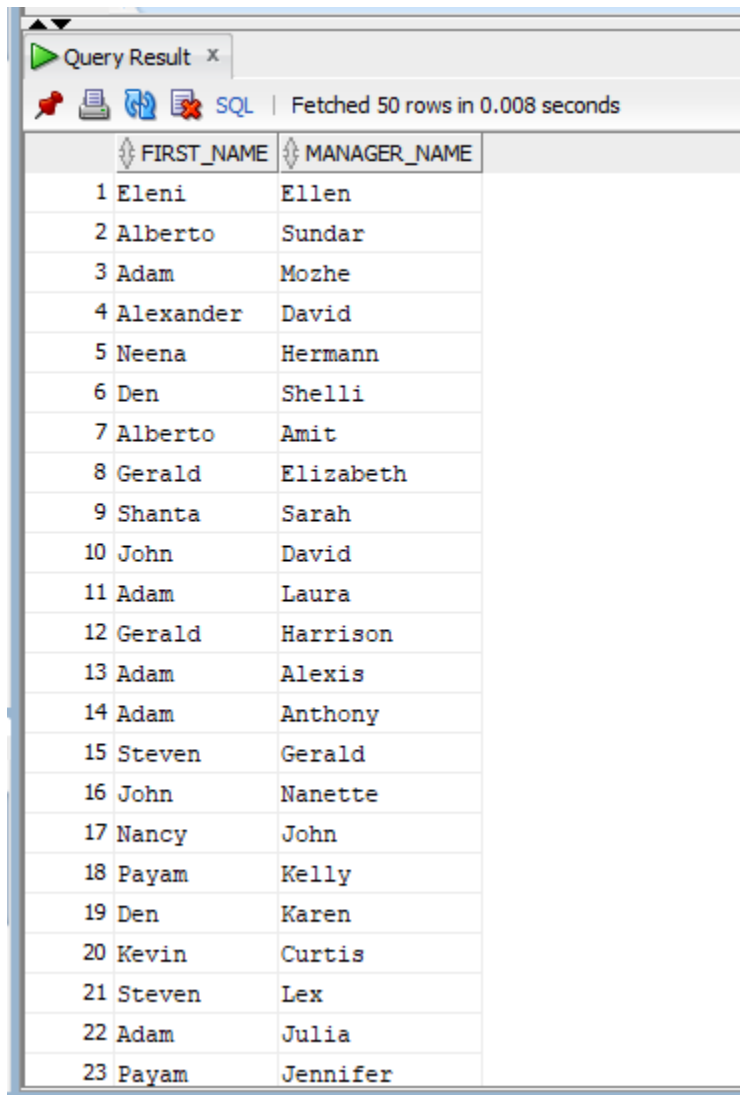
The screenshot shows a 'Query Result' window with a toolbar at the top containing icons for a pin, print, SQL editor, and a close button. The status bar indicates 'All Rows Fetched: 34 in 0.006 seconds'. The query result is displayed in a table with two columns: 'FIRST_NAME' and 'DEPARTMENT_NAME'. The table contains 23 rows of data, all of which are employees from the 'Sales' department. The first names are listed in ascending order of their employee ID.

FIRST_NAME	DEPARTMENT_NAME
1 John	Sales
2 Karen	Sales
3 Alberto	Sales
4 Gerald	Sales
5 Eleni	Sales
6 Peter	Sales
7 David	Sales
8 Peter	Sales
9 Christopher	Sales
10 Nanette	Sales
11 Oliver	Sales
12 Janette	Sales
13 Patrick	Sales
14 Allan	Sales
15 Lindsey	Sales
16 Louise	Sales
17 Sarath	Sales
18 Clara	Sales
19 Danielle	Sales
20 Mattea	Sales
21 David	Sales
22 Sundar	Sales
23 Amit	Sales

The query makes a table with (First_Name) from Employee table and (department name) from department table and joins using LEFT JOIN them in one table where the department_id from the Employee table equals the department_id from the department table in condition of that the department name from the department table equals to Sales.

Query Review 4

```
SELECT e.first_name, m.first_name AS manager_name
FROM employees e
JOIN employees m
ON e.employee_id = m.manager_id;
```



The screenshot shows a 'Query Result' window with a toolbar containing icons for a red pin, a printer, a refresh, a document with a red X, and the text 'SQL | Fetched 50 rows in 0.008 seconds'. The table below has two columns: 'FIRST_NAME' and 'MANAGER_NAME'. It contains 23 rows of data, each with a row number in the first column. The data represents a self-join of the employees table where each employee is paired with their manager.

	FIRST_NAME	MANAGER_NAME
1	Eleni	Ellen
2	Alberto	Sundar
3	Adam	Mozhe
4	Alexander	David
5	Neena	Hermann
6	Den	Shelli
7	Alberto	Amit
8	Gerald	Elizabeth
9	Shanta	Sarah
10	John	David
11	Adam	Laura
12	Gerald	Harrison
13	Adam	Alexis
14	Adam	Anthony
15	Steven	Gerald
16	John	Nanette
17	Nancy	John
18	Payam	Kelly
19	Den	Karen
20	Kevin	Curtis
21	Steven	Lex
22	Adam	Julia
23	Payam	Jennifer

The query makes a table with the (First name) from employee table, (First name) from employee table and names the record as (manager_name) and joins them in condition that the employee_id from employee table equals to manager_id from employees table. This shows an output but logically is incorrect; we are searching in the manager table where manager_id matches the employees_id and on the select

statement, we are printing the managers name under First_name column.
The fix is :

```
SELECT e.first_name, m.first_name AS manager_name
FROM employees e
JOIN employees m
    ON e.manager_id = m.employee_id;
```